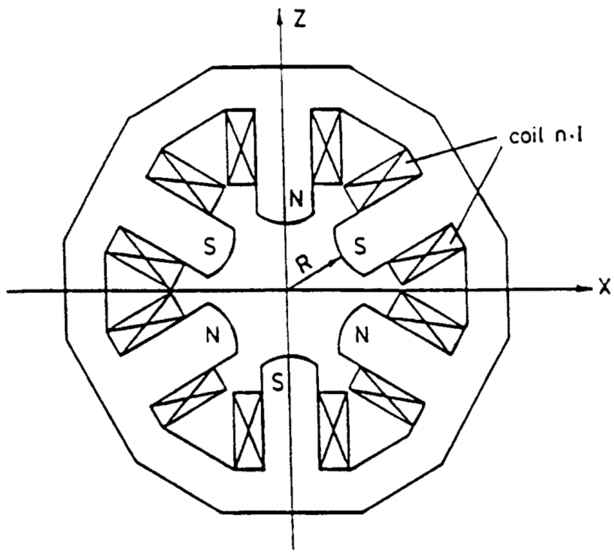
Project 1. Due Tuesday, February 10, 11:59 pm.

1. Implement your favorite Runge-Kutta method (could be as straightforward as Euler's method, although it's a bit boring, or as complicated as RK78 with automatic step control). Any language would do, I would use Python, but the choice is yours. Make sure you actually implement the algorithm, not use the existing algorithm from some library. Verify that your algorithm works by testing it on some simple ODE with known solution, e.g., $y'(t) = y(t)$ or even $y'(t) = t^2$. Ideally, you would want to compare not one but at least two algorithms with the exact solution, e.g., Euler's method and RK4. Add some figures to your report comparing the approximations with the exact trajectory for some initial condition $y(0) = y_0$.



2. A sextupole magnet (see figure) generates a nonlinear field $B_y = \frac{1}{2}g'(x^2 - y^2)$, $B_x = g'xy$, which can be written as gradient of the potential

$$V(x, y) = -\frac{1}{2}g'\left(x^2y - \frac{1}{3}y^3\right).$$

Using Ampere's law, derive the expression for g'. The idea is the same as that for the quadrupole. Hint: take a look at the shape of the magnetic field lines, use it to your advantage. Show a quick diagram of a sextupole with the integration path in your report. If you are stuck, check CERN'94 accelerator school notes. If all else fails, re-derive the expression for the quadrupole, but make sure to explain every step of the derivation in detail.

3. Use Bmad to design an *imaging* system (imaging in both planes) similar to the one we discussed in class: start with 10-GeV protons, guide them through a triplet of quadrupoles. Choose the length of the quadrupoles, the drifts between them, and the quadrupole strength. Quadrupole length could be set to some reasonable fixed number, say, 0.5 m, drifts should be a bit longer, say, 2 to 5 m, but could be fixed too. Hence, your main parameters are the quadrupole strengths, and you can simplify the process by using the apparent symmetry of the problem. Prove to yourself that the system you've created is indeed imaging. Some graphs and tables will help your with this demo: Twiss parameter plots, Twiss parameter evolution table, transfer matrices saved to a file. To use Bmad optimization feature, the easiest way is to start with two triplet cells back to back. Alternatively, you can adjust the quadrupole strengths manually.